

A degree-gap obstruction to joint complete monotonicity

Solution packet for arXiv:2506.08447v2

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Status. Candidate full solution, likely valid, awaiting human review.

1 Source question and answer

Khasnis and Sholapurkar study when

$$\beta(m, n) = \frac{1}{b(m) + a(m)n}, \quad m, n \in \mathbb{Z}_+, \quad (1)$$

is jointly completely monotone [1]. In their Question 4.6 they take

$$a(x) = a_0(x + a_1), \quad b(x) = b_0(x + b_1)(x + b_2)(x + b_3) \quad (2)$$

with positive parameters satisfying $0 < b_1 < a_1 < b_2 < b_3$, and ask whether the answer is yes. The current arXiv version still poses the question on source PDF page 10:

It turns out that

$$\frac{b(x)}{a(x)} = \frac{b_0}{a_0} \left((x + b_1)(x + b_2 + b_3 - a_1) + (b_3 - a_1)(b_2 - a_1) + \frac{\prod_{j=1}^3 (b_j - a_1)}{x + a_1} \right).$$

At this stage, in the proof of Theorem 3.4 we used the fact that $\{t^{km}\}_{m \in \mathbb{Z}_+}$ is a Hausdorff moment sequence. However, a routine calculation suggests that the sequence $\{t^{km^2}\}_{m \in \mathbb{Z}_+}$ is not a Hausdorff moment sequence. Thus the argument in the proof of Theorem 3.4 breaks down in the case where $l < k - 1$. The indication prompts us to pose the following question:

Question 4.6. *Given the net $\left\{ \frac{1}{p(m, n)} \right\}_{m, n \in \mathbb{Z}_+}$ (where p defined in equation (5)) satisfying $0 < b_1 < a_1 < b_2 < b_3$, is the net $\left\{ \frac{1}{p(m, n)} \right\}_{m, n \in \mathbb{Z}_+}$ joint completely monotone?*

As a step forward in dealing with this problem, we provide a family of nets of the form $\left\{ \frac{1}{p(m, n)} \right\}_{m, n \in \mathbb{Z}_+}$ (where p defined in equation (5)) which are not joint completely monotone but the condition $0 < b_1 < a_1 < b_2 < b_3$ is not satisfied.

Example 4.7. *Let $p : \mathbb{R}_+^2 \rightarrow (0, \infty)$ be defined as in equation (5). Let $t_1 = b_1 b_2 b_3$ and $t_2 = (1 + b_1)(1 + b_2)(1 + b_3)$. If*

$$\frac{(t_2 + 3)^2}{\infty} - \frac{(t_1 + 3/2)^2}{\infty} < 1$$

The answer is *no for every choice* in (2). More generally, a polynomial degree gap at least two always rules out joint complete monotonicity. The proof comes from an exact reduction to one-variable Hausdorff moment sequences.

2 Definitions and the explicit vertical measure

For a sequence $u = (u_m)_{m \geq 0}$, write $\Delta u_m = u_{m+1} - u_m$. It is a Hausdorff moment sequence precisely when

$$(-1)^k \Delta^k u_m \geq 0 \quad (m, k \in \mathbb{Z}_+),$$

equivalently when $u_m = \int_0^1 s^m d\eta(s)$ for a finite positive measure η on $[0, 1]$. The two-variable analogue says that a net is jointly completely monotone precisely when it is the moment net of a finite positive measure on $[0, 1]^2$ [2, Proposition 4.6.11]; this is also recalled at the start of the source paper.

The following observation is the key. It is stated for arbitrary positive sequences, so no polynomial structure is used. Put

$$a_m > 0, \quad b_m > 0, \quad r_m = \frac{b_m}{a_m}, \quad h_m(t) = \frac{1}{a_m} t^{r_m-1} \quad (0 < t < 1).$$

Then

$$\int_0^1 t^n h_m(t) dt = \frac{1}{a_m(n+r_m)} = \frac{1}{b_m + a_m n}. \quad (3)$$

3 An exact slice characterization

Theorem 1 (Vertical-slice criterion). *For positive sequences (a_m) and (b_m) , the net*

$$\beta(m, n) = \frac{1}{b_m + a_m n}, \quad m, n \in \mathbb{Z}_+,$$

is jointly completely monotone if and only if, for every $t \in (0, 1)$, the sequence $(h_m(t))_{m \geq 0}$ is a Hausdorff moment sequence.

Proof. Suppose first that β is jointly completely monotone, and let μ be its representing measure on $[0, 1]^2$. For every m , define a measure on $[0, 1]$ by

$$\nu_m(E) = \int_{[0,1] \times E} s^m d\mu(s, t).$$

Its n th moment is $\beta(m, n)$. By (3), the measure $h_m(t) dt$ has exactly the same moments. Hausdorff moment measures are unique: equality on monomials gives equality on polynomials, and Weierstrass approximation gives equality on every continuous function. Hence

$$d\nu_m(t) = h_m(t) dt. \quad (4)$$

For $m, k \geq 0$, the measure

$$\sum_{j=0}^k (-1)^j \binom{k}{j} \nu_{m+j}(E) = \int_{[0,1] \times E} s^m (1-s)^k d\mu(s, t)$$

is positive. By (4), its Lebesgue density is $(-1)^k \Delta^k h_m(t)$, so that density is nonnegative almost everywhere. It is continuous on $(0, 1)$; therefore it is nonnegative everywhere. This holds for all m, k , proving that $(h_m(t))_m$ is a Hausdorff moment sequence for every $t \in (0, 1)$.

Conversely, assume the stated one-variable condition. Let η_t be the unique positive measure on $[0, 1]$ whose moments are $h_m(t)$. We briefly justify that these measures can be integrated without invoking a measurable selection theorem. For every polynomial P , the map $t \mapsto \int P d\eta_t$ is a finite linear combination of the continuous functions $h_m(t)$. Uniform polynomial approximation then makes $t \mapsto \int f d\eta_t$ measurable for every $f \in C[0, 1]$. Indeed, the approximation error is bounded by $\|P - f\|_\infty \eta_t([0, 1]) = \|P - f\|_\infty h_0(t)$.

For $F \in C([0, 1]^2)$ define

$$L(F) = \int_0^1 \left(\int_0^1 F(s, t) d\eta_t(s) \right) dt.$$

Finite sums of products $f(s)g(t)$ approximate F uniformly, which proves measurability of the inner integral. Moreover,

$$\int_0^1 h_0(t) dt = \frac{1}{b_0} < \infty,$$

so L is a bounded positive linear functional. The Riesz representation theorem gives a finite positive measure μ on $[0, 1]^2$ representing L . Finally,

$$\int_{[0,1]^2} s^m t^n d\mu(s, t) = \int_0^1 t^n h_m(t) dt = \beta(m, n)$$

by (3). Thus β is jointly completely monotone. □

4 The degree-gap theorem

Lemma 2 (Two necessary concavity conditions). *If the net in Theorem 1 is jointly completely monotone, then, for every $m \geq 0$,*

$$\Delta^2 r_m \leq 0, \quad a_{m+1}^2 \geq a_m a_{m+2}. \quad (5)$$

Proof. A Hausdorff moment sequence is log-convex by Cauchy–Schwarz. Applying this to $(h_m(t))_m$ from Theorem 1 gives

$$h_{m+1}(t)^2 \leq h_m(t)h_{m+2}(t),$$

or equivalently

$$\frac{a_m a_{m+2}}{a_{m+1}^2} t^{-\Delta^2 r_m} \leq 1 \quad (0 < t < 1). \quad (6)$$

If $\Delta^2 r_m > 0$, the left side tends to infinity as $t \downarrow 0$, a contradiction. Thus $\Delta^2 r_m \leq 0$. Letting $t \uparrow 1$ in (6) gives the second inequality in (5). $\square$

Theorem 3 (Polynomial degree-gap obstruction). *Let a, b be real polynomials which are positive on $\mathbb{Z}_+$ and have positive leading coefficients. If*

$$\deg b - \deg a \geq 2,$$

then the net in (1) is not jointly completely monotone.

Proof. Let $d = \deg b - \deg a \geq 2$ and let $c > 0$ be the ratio of the leading coefficients. The rational function $r(x) = b(x)/a(x)$ satisfies

$$r''(x) = c d(d-1)x^{d-2} + O(x^{d-3}) > 0$$

for all sufficiently large x . (This follows directly by differentiating the asymptotic expansion $r(x) = cx^d(1 + O(x^{-1}))$.) For every sufficiently large integer m ,

$$\Delta^2 r_m = r(m+2) - 2r(m+1) + r(m) = \int_0^1 \int_0^1 r''(m+u+v) du dv > 0.$$

This contradicts Lemma 2. $\square$

Corollary 4 (Full answer to Question 4.6). *For every positive $a_0, a_1, b_0, b_1, b_2, b_3$ —in particular, for every choice satisfying $0 < b_1 < a_1 < b_2 < b_3$ —the net obtained from (2) is not jointly completely monotone.*

Proof. Here $\deg b = 3$ and $\deg a = 1$, so Theorem 3 applies. The root ordering is immaterial. $\square$

5 Scope, checks, and novelty

The result completely answers the concrete ordered-root question and proves a stronger coefficient-free theorem for every polynomial degree gap at least two. The exact slice criterion also identifies what remains in degree gaps zero and one, but it is not presented as a root-by-root solution of the paper’s much broader Problem 1.1.

The sign mechanism is robust: convexity of $r = b/a$ produces $t^{-\Delta^2 r_m}$ in (6), which diverges at zero and cannot be cancelled by the fixed positive coefficient involving a . Thus the proof does not depend on a low-order numerical counterexample or on the ordering of the roots.

A bounded literature check through 12 August 2026 examined the current arXiv v2 record and full text, exact title/id/question/order searches, related rational-function work, and Bhattacharjee–Nailwal’s characterization of completely alternating functions [3]. The latter handles the unprefactored net $1/(\psi(m) + n)$, whereas the present problem has the variable factor $1/a(m)$. No later answer to Question 4.6 or statement of Theorem 3 was found. This is a bounded novelty screen, not an exhaustive bibliographic guarantee.

Human-review recommendation. High priority. Recheck the converse half of Theorem 1 and the exponent sign in (6); these are the only substantive proof points beyond standard Hausdorff theory.

References

- [1] M. N. Khasnis and V. M. Sholapurkar, *Joint Complete Monotonicity of reciprocal of a polynomial in two variables*, arXiv:2506.08447v2 [math.FA], 17 October 2025.
- [2] C. Berg, J. P. R. Christensen, and P. Ressel, *Harmonic Analysis on Semigroups: Theory of Positive Definite and Related Functions*, Graduate Texts in Mathematics 100, Springer, 1984.
- [3] M. Bhattacharjee and R. Nailwal, *A characterization of completely alternating functions*, Journal of Approximation Theory 313 (2026), 106230, doi:10.1016/j.jat.2025.106230.